

NIYOJAN: An Agentic AI-Based Decision Intelligence System for Strategic Demand Planning in Supply Chains

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Abstract:

Demand forecasting is essential in supply chain management, yet most systems provide only numerical predictions with limited decision support. This study presents **NIYOJAN**, an agentic AI-based decision intelligence framework designed to convert forecasts into actionable insights. The system integrates an LSTM-based forecasting model with a deterministic inventory risk evaluation layer and an agentic reasoning module for orchestrating analytical workflows and scenario simulations.

The framework enables what-if analysis by simulating demand variations and computing key inventory metrics such as safety stock and reorder points. A generative AI component generates human-readable insights while maintaining separation from deterministic logic to ensure reliability.

Experimental results demonstrate improved forecasting accuracy compared to traditional methods, along with enhanced interpretability and planning capability. Overall, NIYOJAN bridges the gap between prediction and decision-making, enabling more effective and proactive supply chain planning.

Keywords: *Agentic AI, Demand Forecasting, Decision Intelligence, Supply Chain Optimization, LSTM, Scenario Simulation*

Introduction:

Demand forecasting is a foundational component of supply chain management [1], directly impacting inventory planning, procurement, production scheduling, and customer satisfaction. Inaccurate forecasts often result in stockouts, overstocking, and increased operational costs, making reliable forecasting systems essential for modern businesses.

Traditional forecasting techniques such as moving averages and ARIMA models have been widely used [2] due to their simplicity and interpretability. However, these methods struggle to capture nonlinear patterns and long-term temporal dependencies present in real-world sales data. Recent advancements in machine learning, particularly deep learning models such as Long Short-Term Memory (LSTM) networks, have significantly improved forecasting accuracy [1][3] by modeling sequential dependencies in time-series data.

Despite these advancements, most existing systems remain prediction-centric, providing numerical outputs without meaningful decision support [4]. In real-world scenarios, decision-makers require actionable insights, interpretability, and strategic recommendations rather than raw predictions. Furthermore, many research efforts focus primarily on model performance while overlooking system-level integration [1], explainability, and deployment considerations.

To address these limitations, this study introduces NIYOJAN, an end-to-end decision intelligence framework that extends beyond forecasting. The system integrates predictive modeling with deterministic decision logic, scenario simulation, and agentic AI-based reasoning to convert forecasts into actionable planning strategies. By combining these components, NIYOJAN enables more informed and proactive decision-making in supply chain environments.

Methods and Materials:

System Design and Methodology:

The NIYOJAN framework is designed as a modular system consisting of multiple interconnected layers, including a forecasting engine, decision intelligence layer, scenario simulation module, and agentic reasoning system. These components work together to transform historical data into actionable business insights.

The forecasting component utilizes an LSTM-based model trained on historical sales data. The model processes sequential input data and predicts future demand values, enabling accurate time-series forecasting. A rolling prediction strategy is implemented, allowing the model to generate multi-step forecasts by iteratively using previous predictions as inputs.

Decision Logic and Inventory Risk Evaluation:

To convert forecasts into actionable decisions, a deterministic decision layer is introduced. This layer evaluates inventory risk by comparing predicted demand with current stock levels. The system classifies inventory conditions into high, medium, or low risk categories based on predefined thresholds.

This approach enables clear and interpretable decision-making [6], such as identifying when to restock, hold inventory, or reduce stock levels. Unlike purely predictive systems, this layer provides direct operational guidance.

Dataset Description and Preprocessing:

The dataset used in this study consists of approximately 50,000 weekly retail sales records covering 120 products over a three-year period. Each record includes product identifiers, time indices, and sales quantities.

Data preprocessing steps included handling missing values, removing outliers, and normalizing the data using Min-Max scaling to ensure consistent input ranges. The dataset was split chronologically into training (70%), validation (10%), and testing (20%) sets to preserve temporal dependencies and avoid data leakage.

Scenario Simulation Engine:

The system incorporates a deterministic scenario simulation engine to evaluate inventory outcomes under varying demand conditions. This module enables planners to perform counterfactual analysis by modifying forecast values and analyzing their impact on inventory risk.

Demand variation scenarios are computed as:

$$\text{NewDemand} = \text{Forecast} \times (1 + p/100) \quad (1)$$

where p represents the percentage change applied to the baseline forecast.

Inventory planning metrics are calculated using standard supply chain formulations:

$$\text{SafetyStock} = Z \times \sigma \times \sqrt{\text{LeadTime}} \quad (2)$$

$$\text{ROP} = (\text{Demand} \times \text{LeadTime}) + \text{SafetyStock} \quad (3)$$

$$\text{Inventory Gap} = \text{ROP} - \text{CurrentStock} \quad (4)$$

where Z represents the service level factor, σ is the standard deviation of demand, and LeadTime denotes the replenishment lead time.

These deterministic computations enable accurate evaluation of inventory risks under simulated demand conditions, allowing planners to identify potential stock shortages or excess inventory and make informed decisions.

Agentic AI Workflow:

The agentic reasoning layer acts as the intelligence core of the system. It orchestrates multiple analytical processes, including data retrieval, scenario simulation, and strategic planning.

The workflow begins with intent detection and contextual analysis, followed by routing tasks to appropriate analytical modules. The system then performs simulations and generates structured insights. Importantly, deterministic computations are kept separate from generative AI reasoning to ensure reliability and prevent incorrect outputs.

Implementation and Deployment:

The forecasting model is implemented using TensorFlow and Keras, with two stacked LSTM layers containing 64 hidden units each, followed by a dense output layer. The model is trained using the Adam optimizer with a learning rate of 0.001 and Mean Squared Error (MSE) as the loss function.

The entire system is deployed using Docker containers on an AWS EC2 instance. Backend services are built using FastAPI, exposing functionalities through REST APIs. This architecture ensures scalability, portability, and efficient system operation.

Results:

The performance of the NIYOJAN framework was evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), which are standard metrics for time-series forecasting.

Table 1. Comparison of Forecasting Models

Model	MAE	RMSE
PROPHET	12.8	16.5
ARIMA	14.2	19.1
LSTM (NIYOJAN)	9.8	12.4

The results demonstrate that the LSTM-based NIYOJAN model outperforms traditional statistical approaches [1][5]. Specifically, the proposed model achieves approximately 31%

lower MAE compared to ARIMA and 23% lower MAE compared to Prophet, highlighting its effectiveness in capturing nonlinear patterns and temporal dependencies in demand data.

Beyond improved forecasting accuracy, the framework enhances decision-making by transforming predictions into actionable insights. The decision intelligence layer identifies inventory risks, while the scenario simulation module enables evaluation of varying demand conditions for informed planning.

To further assess model performance, a visual comparison between actual and predicted demand values is presented in Figure 1. The results show strong alignment with real demand trends, indicating that the model effectively captures both short-term fluctuations and long-term patterns.

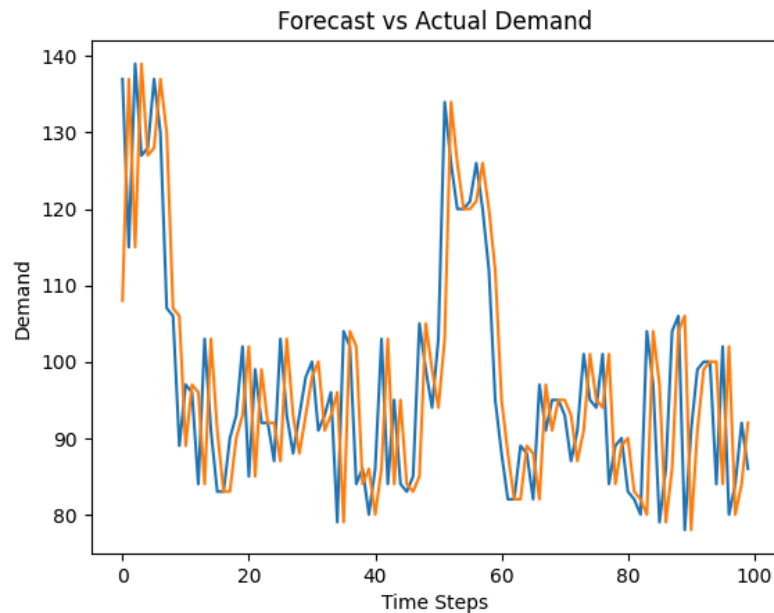


Figure 1. Forecast vs Actual Demand showing the alignment between predicted and actual values over time.

Additionally, the distribution of residual errors is illustrated in Figure 2, showing that most prediction errors are centered around zero, indicating low bias.

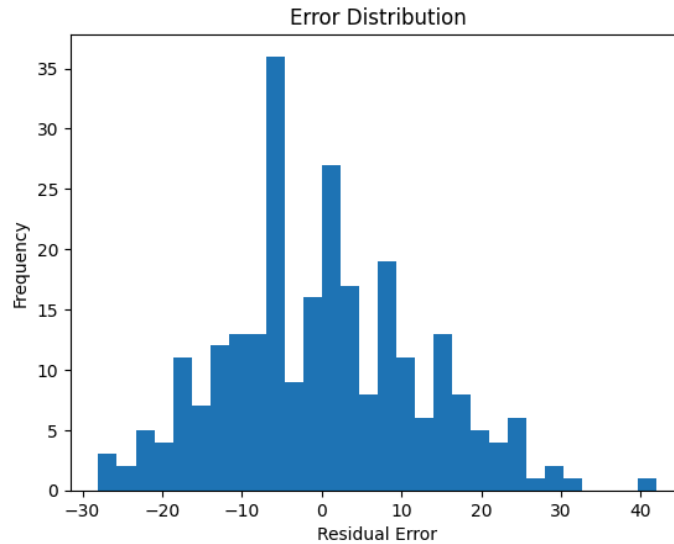


Figure 2. Distribution of residual errors indicating prediction accuracy and model consistency.

This improvement directly reduces inventory uncertainty and enables more efficient supply chain planning in practical scenarios. Overall, the NIYOJAN framework not only improves predictive performance but also enhances interpretability and supports proactive, data-driven decision-making.

System Architecture:

The overall system architecture is illustrated in Figure 3. The architecture consists of the user interface layer, backend API layer, forecasting engine, decision intelligence layer, and the agentic AI intelligence hub responsible for strategic planning.

These interconnected components form a continuous pipeline that transforms raw historical data into actionable business intelligence. The foundational layers handle data ingestion, LSTM-based forecasting, and deterministic risk classification. The agentic intelligence hub functions as the cognitive core of the system by orchestrating retrieval-based knowledge access, scenario simulations, and structured reasoning workflows.

By separating deterministic computations from generative AI reasoning, the system ensures that all analytical outputs remain mathematically consistent while still providing human-readable strategic insights. This design minimizes the risk of unreliable outputs and enhances trust in decision-making processes.

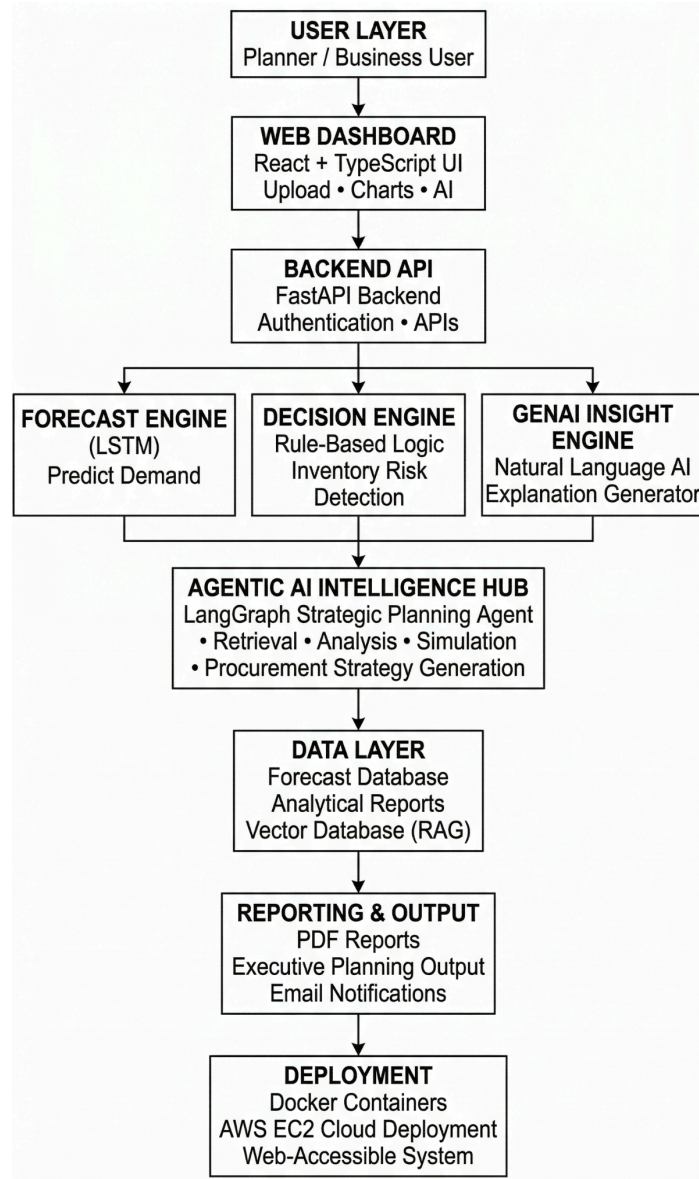


Figure 3. System architecture of the proposed NIYOJAN framework integrating forecasting, decision intelligence, and agentic AI-based strategic planning.

Discussion:

The findings of this study demonstrate that integrating predictive modeling with decision intelligence significantly enhances the practical utility of demand forecasting systems [4]. While traditional approaches focus primarily on prediction accuracy [2][5], NIYOJAN extends this capability by incorporating deterministic reasoning and agentic AI-based planning.

One of the key strengths of the proposed framework is the clear separation between deterministic computations and generative AI reasoning. This design ensures that critical calculations, such as inventory planning metrics, remain accurate and reliable, while the generative module enhances interpretability by providing human-readable insights.

The inclusion of scenario simulation further strengthens the system [6] by enabling users to analyze different demand conditions and assess potential risks. This capability is particularly valuable in dynamic supply chain environments where uncertainty is high.

However, the study has certain limitations. The dataset used is limited in scope and may not capture all real-world complexities. Additionally, the system assumes relatively stable lead times, which may not always hold in practice. Future work can address these limitations by incorporating real-time data streams, adaptive decision thresholds, and reinforcement learning-based optimization techniques.

Conclusion:

This study introduced NIYOJAN, an agentic AI-driven decision intelligence framework designed to enhance strategic demand planning. By integrating LSTM-based forecasting, deterministic decision logic, scenario simulation, and agentic reasoning, the system transforms numerical predictions into actionable insights.

The results demonstrate that the proposed approach not only improves forecasting accuracy but also enhances interpretability and decision-making capability. This makes NIYOJAN more suitable for real-world applications compared to traditional prediction-centric systems.

Future research will focus on extending the framework to multi-agent systems, real-time analytics, and advanced optimization methods to further improve its effectiveness. This positions NIYOJAN as a scalable and practical solution for next-generation intelligent supply chain systems.

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Conflict of Interest Statement:

The authors have no conflict of interest to disclose.

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